

Symptom: Client shows no IP address or an APIPA address (169.254.x.x)

A Windows or Linux client that fails to receive a DHCP offer will typically self-assign an Automatic Private IP Addressing (APIPA) address in the 169.254.0.0/16 range. This is a strong signal that no DHCP OFFER was received within the client's timeout window, not that the client itself is misconfigured.

Cause 1: DHCP Pool Exhaustion

If every address in a DHCP scope is currently leased, new clients requesting an address will not receive an offer. This is common in environments with high device turnover (guest networks, BYOD deployments) where lease times are set too long relative to how quickly devices join and leave the network.

Diagnostic steps:

- Check the scope's current utilization on the DHCP server (most platforms expose this as "addresses leased" vs. "pool size")
- Look for stale leases — devices that received an address but disconnected without releasing it, which can hold addresses until the lease naturally expires
- Consider shortening lease duration on high-churn networks, or expanding the scope's address range if physically possible

Cause 2: Relay Agent Misconfiguration

When the DHCP server and client sit on different subnets or VLANs, a relay agent (typically the client's default gateway router) is responsible for forwarding DHCP broadcast traffic to the server as a unicast request. If the relay agent is missing, misconfigured, or pointing to the wrong server address, DHCP requests never leave the client's local segment.

Diagnostic steps:

- Confirm the relay agent (ip helper-address or equivalent) is configured on the client-facing VLAN interface, pointing to the correct DHCP server IP
- Verify there's no access control rule blocking UDP ports 67/68 between the relay agent and the DHCP server
- If multiple DHCP servers exist, confirm the relay agent is forwarding to all intended servers, not just one

Cause 3: Address Conflicts

Occasionally, two devices end up using the same IP address — often because a static IP was manually assigned within the DHCP scope's range, or because a lease wasn't properly released before being reassigned. Most DHCP servers perform a conflict check (via ping or gratuitous ARP) before finalizing an offer; if a conflict is detected, the address is pulled from the available pool until manually resolved.

Diagnostic steps:

- Review the DHCP server's conflict log, if available
- Confirm no statically-assigned devices fall within the dynamic pool's address range
- If conflicts recur, consider excluding a reserved range from the dynamic pool specifically for static assignments

Cause 4: Physical or VLAN Connectivity

Before assuming a DHCP-specific issue, confirm basic connectivity: the client's switchport is up, assigned to the correct VLAN, and not blocked by a security feature (port security, DHCP snooping trust settings) that could silently drop DHCP traffic.

Symptom: Client receives an IP address, but it's from the wrong subnet or VLAN

Overview

A client obtains an address successfully — no timeout, no APIPA fallback — but the assigned address doesn't match the subnet it should be on. This usually points to a DHCP scope selection or relay problem, not a failure of DHCP itself.

Cause 1: Multiple DHCP Servers Responding

If more than one DHCP server is reachable from the client's broadcast domain (a rogue server, a misconfigured test server left running, or a second legitimate server with overlapping scope), the client accepts whichever DHCPOFFER arrives first — which may not be from the server you intended.

Diagnostic steps:

- Use a packet capture on the client's segment during a DHCP request to see how many DHCPOFFER responses come back, and from which source IPs
- Audit the network for unauthorized or forgotten DHCP servers (a common cause: a consumer router plugged into the LAN with its own DHCP server still enabled)
- Enable DHCP snooping (or equivalent) on access switches to restrict which ports are trusted to send DHCP server traffic

Cause 2: Relay Agent Pointing to the Wrong Scope

If a relay agent is configured but forwarding to a DHCP server's scope intended for a different VLAN, clients receive valid-looking leases from the wrong address range entirely.

Diagnostic steps:

- Confirm the relay agent's configured helper address matches the DHCP server instance/scope intended for that specific VLAN

- On the DHCP server, verify each scope's associated VLAN/subnet metadata is correct and hasn't been accidentally duplicated or misassigned during a recent change
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Symptom: DHCP works intermittently — some devices get addresses, others time out

Overview

Unlike total DHCP failure, intermittent success suggests a timing, capacity, or partial-connectivity issue rather than a complete misconfiguration — DHCP is functioning, but not reliably for every request.

Cause 1: DHCP Server Under Load

A DHCP server handling a burst of simultaneous requests (e.g., after a power outage when hundreds of devices reconnect at once) may not respond to every request within a client's retry window, causing some devices to fail while others succeed on a later retry.

Diagnostic steps:

- Check the DHCP server's CPU/memory utilization during the affected window
- Review server logs for dropped or delayed requests correlating with the timing of reported failures
- Consider whether request volume spikes (mass reconnects, VDI boot storms) exceed the server's configured capacity

Cause 2: Intermittent Relay Path Connectivity

If the path between a relay agent and the DHCP server is unstable (flapping link, marginal wireless backhaul, an oversubscribed uplink), some DHCP packets are lost in transit while others get through — appearing as random client-side failures.

Diagnostic steps:

- Check interface error counters and link stability logs on the path between relay agent and server
 - Test with sustained ping/traceroute to the DHCP server from the relay agent's source interface to look for intermittent loss
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Symptom: DHCP requests succeed, but clients lose connectivity shortly after

Overview

The client receives a valid, correctly-scoped address, but connectivity fails soon after — this points away from DHCP itself and toward what happens immediately after lease assignment: gateway reachability, DNS options, or a very short lease time.

Cause 1: Lease Time Set Too Short

If a DHCP scope's lease duration is misconfigured to an unusually short value (minutes instead of hours/days), clients may appear to lose connectivity simply because their lease expired and renewal failed or was delayed.

Diagnostic steps:

- Check the configured lease time on the scope — anything under roughly 30-60 minutes for a standard wired/wireless client network is unusually aggressive and worth questioning
- Correlate the timing of reported disconnects against the lease duration to see if they line up

Cause 2: Incorrect DHCP Options (Gateway/DNS)

A client can receive a technically valid IP address but still fail to communicate if the DHCP scope's option set — default gateway (option 3) or DNS servers (option 6) — is wrong, blank, or pointing to a decommissioned device.

Diagnostic steps:

- Inspect the client's received DHCP options (ipconfig /all on Windows, dhclient lease file on Linux) to confirm gateway and DNS values are correct and reachable
- Compare against the DHCP scope's configured options on the server to catch a typo or stale entry

Here are the two, same format:

Symptom: Legitimate DHCP server traffic is being dropped after enabling port security features

Overview

After enabling DHCP snooping (or a similar port-security feature) on access switches, DHCP requests stop being answered — even though the DHCP server itself hasn't changed. This is a common side effect of security hardening: the feature designed to block *rogue* DHCP servers can also silently block a *legitimate* one if the trust configuration doesn't match the real topology.

Cause 1: DHCP Server Port Not Marked as Trusted

DHCP snooping works by only allowing DHCP OFFER and DHCP ACK messages (server-to-client responses) to pass through ports explicitly marked "trusted." Every other port is treated as untrusted and only allowed to send client-side messages (DHCP DISCOVER, DHCP REQUEST). If the uplink port toward the actual DHCP server isn't marked trusted, its responses get silently dropped.

Diagnostic steps:

- Check the DHCP snooping trust configuration on every switch between the client and the DHCP server — the port facing the DHCP server (or facing the relay agent/router, if the server is elsewhere) must be trusted
- Confirm trust settings are applied consistently across the full path if there are multiple switches in a chain, not just the switch directly connected to the client
- Review switch logs for DHCP snooping violation messages, which typically name the exact port and packet type that was dropped

Cause 2: Rate Limiting Threshold Too Aggressive

DHCP snooping and related port-security features often include a rate limit on DHCP packets per port, intended to prevent a single misbehaving or malicious device from flooding the network. If this threshold is set too low, a burst of legitimate requests (e.g., many devices rebooting after a power event) can get throttled or dropped.

Diagnostic steps:

- Check the configured DHCP rate limit per port and compare it against expected peak request volume (e.g., after a scheduled reboot or mass reconnect)
- Look for rate-limit violation counters incrementing on affected ports during the reported outage window

Symptom: Device shows a stale or invalid IP address after being moved to a different port, VLAN, or network segment

Overview

A client that previously had working connectivity keeps its old IP address configuration after being physically moved (different switch port, different VLAN, different site) — and that address no longer matches the new segment. This usually isn't a DHCP server problem at all; it's a client-side caching behavior interacting badly with a topology change.

Cause 1: Client Attempting to Renew Instead of Requesting Fresh

Most DHCP clients try to renew their existing lease first (a unicast DHCPREQUEST to the last known server) before falling back to a full broadcast DHCPDISCOVER. If the client has moved to a segment where the original DHCP server is unreachable, the renewal silently fails, but the client may continue using the old (now invalid) address until the lease fully expires or the OS forces a rebind.

Diagnostic steps:

- Check the client's current lease details and compare the lease's originating server/scope against what's expected for its current location

- Force a full release and renew on the client (ipconfig /release + /renew on Windows, or the equivalent dhclient -r / restart on Linux) to confirm a fresh request resolves the issue — if it does, this confirms stale-lease caching was the cause, not a server-side problem

Cause 2: Switch Port Still Assigned to the Old VLAN

If the physical move wasn't accompanied by a corresponding VLAN reassignment on the new switch port (e.g., the port is still configured for the old VLAN by default or by oversight), the client is technically still on its original broadcast domain from the network's perspective, and will continue receiving addresses from the original scope rather than the one intended for its new location.

Diagnostic steps:

- Verify the actual VLAN assignment of the switch port the device is now connected to, and confirm it matches the intended segment
 - Compare this against the port's expected VLAN per network documentation or the port's naming/description convention, if one exists
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